Q18: LLMs vs. Brains

What the analogy gets right and wrong

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The question

Student question (#06, #11)
How similar is LLM "thinking" to human thinking? How similar are AI systems to biological brains, and could we build AI from brain structure?

Separate three claims

1. **Behavioral** similarity (same outputs).
2. **Representational** similarity (same internal codes).
3. **Mechanistic** similarity (same process).

Often conflated; very different evidence.

Where the analogy holds

- Distributed representations + superposition (both).
- Hierarchical features; model activations *predict* sensory cortex.
- Learning from experience (adapt connection strengths).

Where it breaks

- **Learning rule**: backprop is biologically implausible.
- **Substrate**: spiking, stochastic, $\sim$20 W brain vs clocked GPUs.
- **Objective**: next-token text vs embodied, evolved, reward-driven.
- **Architecture**: attention/KV-cache has no cortical analog.

Build AI from brain structure?

- Transfers that worked were **abstractions** (distributed codes, convolution, attention).
- Literal spiking/connectome models have *not* matched deep learning on tasks.
- Flow reverses: deep nets are now a leading *model* of sensory cortex.

"Copy the connectome and run it" is not a proven route to capability.

Takeaway

Shared: distributed, hierarchical, experience-shaped representations.
Different: learning rule, substrate, objective, architecture.

Behavioral resemblance $\neq$ mechanistic identity.